

LUCIA GHIRETTI

PhD student - Dept. of Food and Drug, University of Parma

CONTACT

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SKILLS

- Multi-omics integration
- (Poly)phenols
- Cardiometabolic diseases
- R Studio
- Basic HPC usage
- Basic Bash / Command line

NETWORKS

- International Society for Computational Biology (ISCB)
- ISCB Regional Study Group Italy (RSG-Italy)
- ISCB Student Council, BioinfoBridges
- NuGO Bioinformatics & Systems Biology Working Group,

RESEARCH

My research focuses on multi-omics integration approaches to investigate inter-individual variability in the processing of dietary (poly)phenols, key bioactive compounds, with the aim of understanding their roles in cardiometabolic diseases and fostering precision nutrition.

EXPERIENCES

PhD student

University of Parma • Nov 2024 - Present

2nd year PhD Student as Bioinformatician at the Human Nutrition Unit, mainly focusing on the development of precision nutrition strategies through multi-omics, data science and ML approaches.

Associate analytical technician

Chiesi Farmaceutici • Jul - Nov 2024

Analytical technician supporting dry powder inhalation product development.

Master student traineeship

University of Parma • Mar - Jul 2024

Flavonoids metabolism - Mass spectrometry for the identification and quantification of flavonoids in urine samples, in order to identify different metabolotypes and classify donors through them.

Master student traineeship

Wageningen University & Research • Jun - Dec 2023

Colorectal cancer - Potential role of phenyl- γ -valerolactones (PVLs), specific gut microbial metabolites (GMMs), in the prevention of CRC using healthy and cancerous human apical-out (AO) colon organoids.

EDUCATION

Master degree in Genomic, Molecular and Industrial Biotechnology

University of Parma • Oct 2021 - Apr 2024

(Poly)phenol catabolism - Static and dynamic in vitro colonic models reveal the spatiotemporal production of flavan-3-ol catabolites.

Bachelor degree in Biotechnology

University of Parma • Oct 2018 - Jul 2021

Mitochondrial diseases - Yeast *Saccharomyces cerevisiae* as a model for the study of trans mutation in the WARS2 gene.

RELEVANT PUBLICATIONS AND CONFERENCES

- Ghiretti, L., Treccani, M., Milenkovic, D., Dei Cas, A., Del Rio, D., Mena, P. (2026). Exploring multi-omics approaches in (poly)phenol research for precision nutrition. *Food & function*, 17(16), 7030-7053. <https://doi.org/10.1039/d6fo02312f>.
- Treccani, M., Ghiretti, L., Tosi, N., Mignogna, C., Minihane, A. M., Barili, V., Del Rio, D., Martorana, D., & Mena, P. (2025). A computational reference for human genomics analysis in (poly)phenol research. *Food & function*, 16(24), 9408-9421. <https://doi.org/10.1039/d5fo03227j>
- Ma, Y., Ghiretti, L., Castellone, V., Mena, P., & Rubert, J. (2025). Static and dynamic in vitro colonic models reveal the spatiotemporal production of flavan-3-ol catabolites. *Free radical biology & medicine*, 227, 582-592. <https://doi.org/10.1016/j.freeradbiomed.2024.12.034>
- Attended ISMB/ECCB 2025 Conference - Poster presentation - Integrative multi-omics analysis to explore the molecular impact of dietary (poly)phenols on humans - Poster presentation award.
- Attended NuGO 2026 Conference - Oral presentation - Multi-omics integration explains inter-individual variability in dietary (poly)phenol metabolism.